

Pelvic Circumferential Compression Binder

Guideline Title: Pelvic Circumferential
Compression Binder

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3203-PR027

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CCCXLII. **PURPOSE:** To provide circumferential compression to reduce and stabilize a suspected pelvic ring disruption. The potential benefits of a pelvic binder include lower mortality rate, reduce blood loss, and aid in pain management.

CCCXLIII. **DEFINITIONS:** None

CCCXLIV. **DEPARTMENT GUIDELINE:**

A. Equipment

- i. Appropriately sized commercial pelvic device "PelvicBinder"
 1. Adult
 2. Pediatric: hip circumference range 14-32 inches **-OR-**
- ii. Standard bed sheet

B. Indications

- i. Suspected or proven unstable pelvic fracture demonstrated on radiograph
- ii. Mechanism of injury consistent with possible pelvic fracture with physical exam
 1. Abrasions and contusions around the pelvic area
 2. Hematoma above the inguinal ligament, to the scrotum, or the thigh
- iii. Mechanism of injury consistent with possible pelvic fracture with hemorrhagic shock without another source of hemorrhage.

C. Contraindications

- i. There are no reported contraindications to using pelvic circumferential compression.

D. Procedure

- i. Gather equipment and supplies. Anticipate need for complete spinal immobilization.
- ii. Consider placing device on patient outside of the EMS rescue truck, on a stretcher, with patient elevated at waist height.
- iii. Assess pelvic area, distal circulation, sensation, and motor function of lower extremities.
- iv. Remove any clothing/objects from pelvic area. In males, ensure genitalia are elevated out of groin area/not between legs. Unclick the backboard straps.
- v. Consider utilizing the 'lift and slide' under method or the 'sawing' method with 3 providers to place device.
- vi. If using PelvicBinder commercial device
 1. Place the pelvic binder beneath the patient and center at the level of trochanters.
 2. Bring the sewed end of the binder over the front of pelvis first, then the loose end
 3. Place the loose end of the plates on the cloth material.
 4. Use the gap with of the two ends as a cutting guide.
"Measure twice, cut once."
 5. Cut the excess material on the medial side of the plates.
 6. After cutting you should have a 4-5-inch gap between the two plates.
 7. Attach the loose end of the binder plates to the belts.
 8. Pull the handle with cord to achieve pelvic reduction. The cord will lock into place at the point you stop pulling.
 9. Secure the loose cord into the cord holder.
 10. If the binder needs to be loosened, pull slight tension on the cord and handle and lock into place.
 11. PelvicBinder commercial device is radiolucent, MRI compatible, & latex free.
- vii. If using Pediatric PelvicBinder
 1. It is half the width and length of standard PelvicBinder
- viii. If using sheet
 1. Fold the sheet smoothly (do not roll the sheet).
 2. Place the sheet under the patient's pelvis so it is centered over the greater trochanters, where the head of the femur attaches to the pelvis.
 3. Wrap and twist the two running ends of the sheet around the patient's pelvis.
 4. Once tightened, cross the running ends and tie/clamp them to maintain tension.
- ix. Reassess distal circulation, sensation, and motor function of lower extremities. Assess for any blood pooling in groin region.
- x. If mobility of the lower extremities is not limited by injuries or severe discomfort, consider binding the feet together with tape or sheet to maintain pelvic anatomical position. Binders can work better if the

lower extremities are also kept close together. Sometimes fractures make this impractical or impossible.

- xi. Re-click backboard straps if initially unclicke to secure patient to board.
- xii. Avoid repeated assessments by manual compression of an unstable pelvis.

CCCXLV. **DOCUMENTATION:** EMS Charts

CCCXLVI. **REFERENCES:**

- A. PelvicBinder Medical Products: <https://www.pelvicbinder.com>
- B. Black, S.R., Sathy, A.K., Jo, C., Wiley, M.R., Minei, J.P., & Starr, A.J. (2016). Improved survival after pelvic fracture: 13-Year experience at a single trauma center using a multidisciplinary institutional protocol. *J Orthop Trauma*, 30(1), 22-8. doi: 10.1097/BOT.0000000000000443. PMID: 26360539.